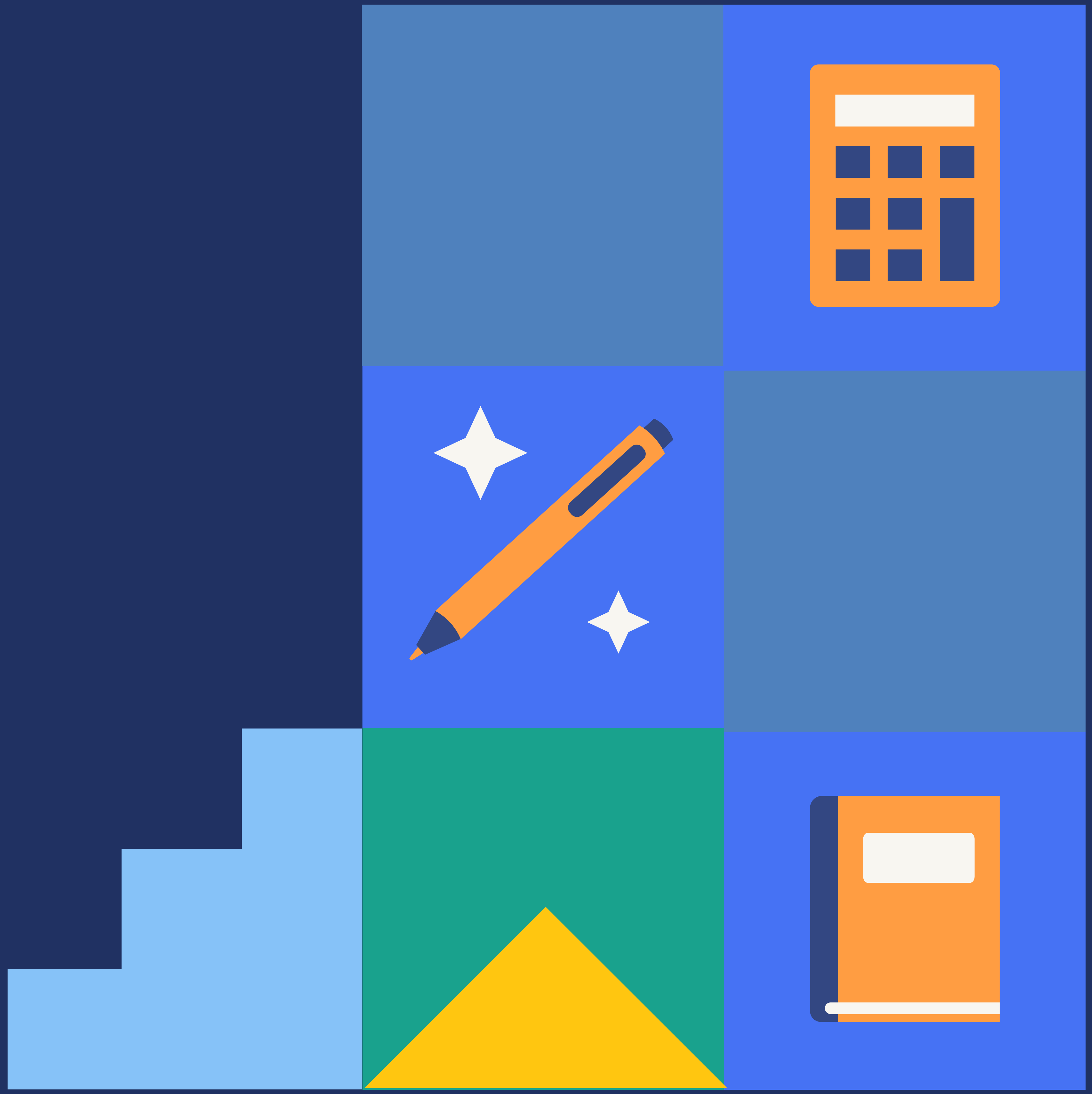
Our Team

Ronit Naik

Mark Lopes

Rohit Patra

Vivian Ludrick



Research Paper

TITLE

An automated approach to predict diabetic patients using KNN imputation and effective data mining techniques

AUTHORS

- Abdulaziz Altamimi
- Aisha Ahmed Alarfaj
- Muhammad Umer
- Ebtisam Abdullah Alabdulqader
- Shtwai Alsubai
- Tai-hoon Kim
- Imran Ashraf

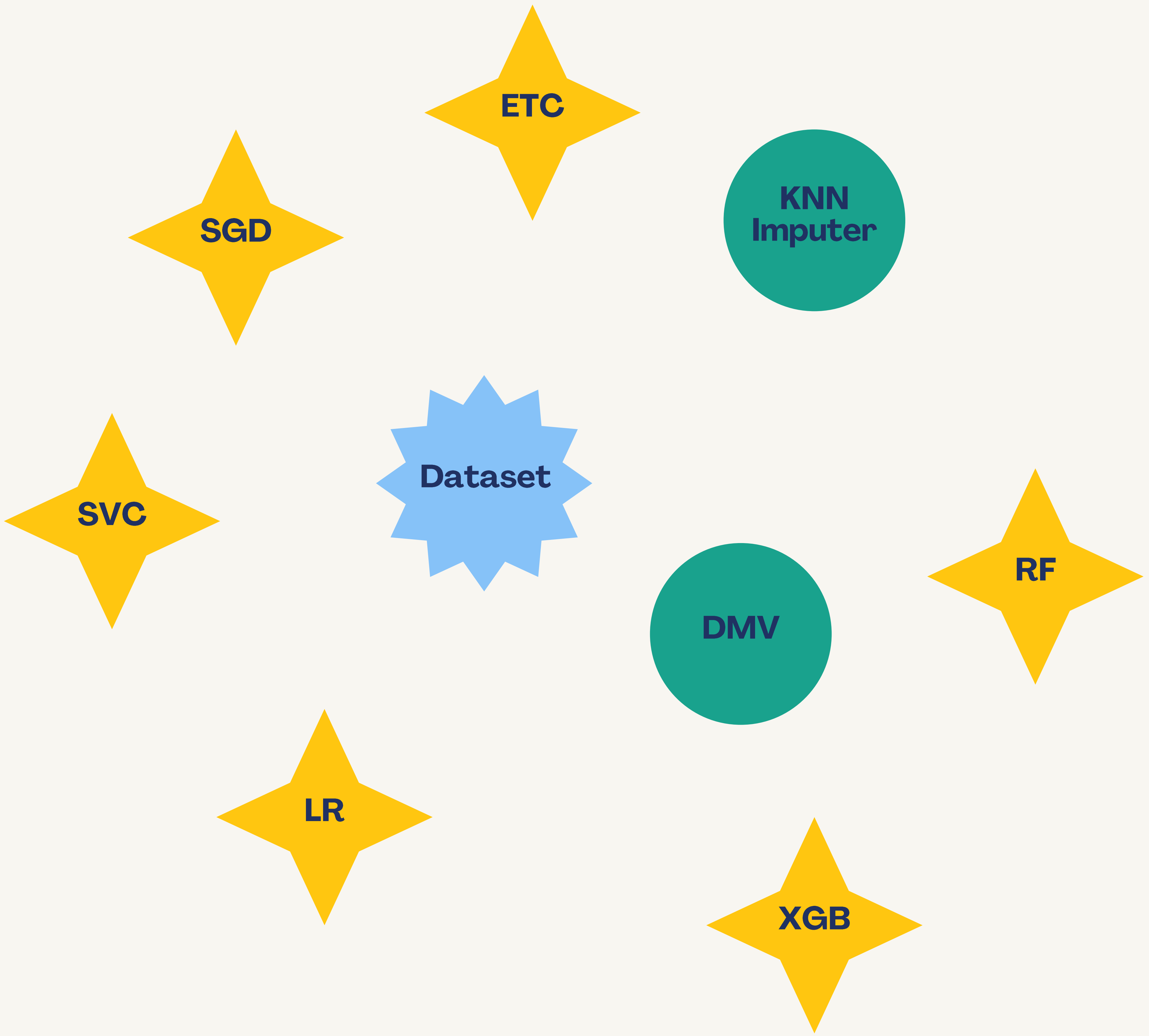
Materials and Methods

Dataset

Models Used for Diabetes Prediction

Data Preprocessing

Materials and Methods



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- Kaggle *(PIMA Indians Diabetes Dataset)*
- It contains 768 records of female patients aged 21 and above
- 268 diabetic patients and 500 non-diabetic patients.
- Multiple features like BMI, Glucose level, Blood Pressure, Skinfold thickness, and Insulin were included, but missing values were found in key attributes.

ETC

SGD

SVC

LR

XGB

RF

DMV

KNN Imputer

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DMV

KNN Imputer

- K-Nearest Neighbors (KNN) Imputation was used to handle missing values in the dataset.
- Rows with missing values can be removed entirely, reducing the dataset size.

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Models Used for Diabetes Prediction

- Logistic regression (LR) is a statistical model that uses a logistic function to model a binary dependent variable (in this case, diabetic or non-diabetic).
- Stochastic Gradient Descent (SGD) is an iterative optimization algorithm used to minimize the cost function of models like Support Vector Machines (SVM) or Logistic Regression.
- A support vector classifier (SVC) separates classes by finding the best hyperplane that maximizes the margin between data points of different classes.

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- Random Forest (RF) is an ensemble learning method that builds multiple decision trees and aggregates their results to produce a more robust model.
- Extra Trees Classifier (ETC) (Extremely Randomized Trees) is similar to Random Forest, but it introduces more randomness during the training process.
- XGBoost(XGB) is a popular and powerful gradient boosting algorithm that sequentially builds models to correct errors made by the previous models.

Dataset

DMV

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Data Preprocessing

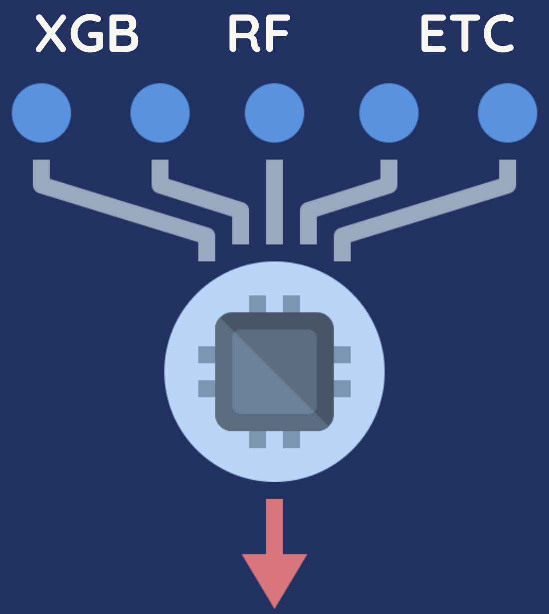
Proposed Solution

Diabetes Dataset

Proposed Solution

Diabetes Dataset

KNN Imputation



Accuracy
Precision
Recall
F1-Score
MCC

Proposed Solution

Algorithm Ensembling XGB, RF, and ETC for diabetes classifcation

Model Training:
Train model_XGB on $(X^*_{\text{train}}, y_{\text{train}})$
Train model_RF on $(X^*_{\text{train}}, y_{\text{train}})$
Train model_ETC on $(X^*_{\text{train}}, y_{\text{train}})$
Model Prediction on Validation Set:
Predict probabilities on X_{val} using model_XGB

$$P_{\text{XGB}} \leftarrow \text{model\_XGB.predict\_proba}(X_{\text{val}})[:, 1]$$

Predict probabilities on X_{val} using model_RF

$$P_{\text{RF}} \leftarrow \text{model\_RF.predict\_proba}(X_{\text{val}})[:, 1]$$

Predict probabilities on X_{val} using model_ETC

$$P_{\text{ETC}} \leftarrow \text{model\_ETC.predict\_proba}(X_{\text{val}})[:, 1]$$

Ensemble Predictions:
Combine the predictions using the average method

$$P_{\text{ensemble}} \leftarrow \frac{P_{\text{XGB}} + P_{\text{RF}} + P_{\text{ETC}}}{3}$$

Determine Optimal Threshold:
Find the optimal threshold τ that maximizes the F1 score on P_{ensemble} and y_{val}
Evaluate Ensemble Model on Validation Set:
Generate binary predictions using τ

$$\hat{y}_{\text{val}} \leftarrow (P_{\text{ensemble}} \geq \tau)$$

Calculate accuracy, precision, recall, and F1 score for $\hat{y}_{\text{val}}$
Final Predictions on Test Set:
Predict probabilities on X^*_{test} using model_XGB

$$P_{\text{XGB\_test}} \leftarrow \text{model\_XGB.predict\_proba}(X^*_{\text{test}})[:, 1]$$

Predict probabilities on X^*_{test} using model_RF

$$P_{\text{RF\_test}} \leftarrow \text{model\_RF.predict\_proba}(X^*_{\text{test}})[:, 1]$$

Predict probabilities on X^*_{test} using model_ETC

$$P_{\text{ETC\_test}} \leftarrow \text{model\_ETC.predict\_proba}(X^*_{\text{test}})[:, 1]$$

Combine the predictions using the average method

$$P_{\text{ensemble\_test}} \leftarrow \frac{P_{\text{XGB\_test}} + P_{\text{RF\_test}} + P_{\text{ETC\_test}}}{3}$$

Generate final binary predictions using τ

$$y_{\text{pred}} \leftarrow (P_{\text{ensemble\_test}} \geq \tau)$$

Output Predictions:
Return y_{pred} for X_{test}

Proposed Solution

Algorithm Ensembling XGB, RF, and ETC for diabetes classifcation

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Predict probabilities on X_{val} using model_XGB

$$P_{\text{XGB}} \leftarrow \text{model\_XGB.predict\_proba}(X_{\text{val}})[:, 1]$$

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Predict probabilities on X_{val} using model_ETC

$$P_{\text{ETC}} \leftarrow \text{model\_ETC.predict\_proba}(X_{\text{val}})[:, 1]$$

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$$\hat{y}_{\text{val}} \leftarrow (P_{\text{ensemble}} \geq \tau)$$

Calculate accuracy, precision, recall, and F1 score for $\hat{y}_{\text{val}}$

Final Predictions on Test Set:

Predict probabilities on X^*_{test} using model_XGB

$$P_{\text{XGB\_test}} \leftarrow \text{model\_XGB.predict\_proba}(X^*_{\text{test}})[:, 1]$$

Predict probabilities on X^*_{test} using model_RF

$$P_{\text{RF\_test}} \leftarrow \text{model\_RF.predict\_proba}(X^*_{\text{test}})[:, 1]$$

Predict probabilities on X^*_{test} using model_ETC

$$P_{\text{ETC\_test}} \leftarrow \text{model\_ETC.predict\_proba}(X^*_{\text{test}})[:, 1]$$

Combine the predictions using the average method

$$P_{\text{ensemble\_test}} \leftarrow \frac{P_{\text{XGB\_test}} + P_{\text{RF\_test}} + P_{\text{ETC\_test}}}{3}$$

Generate final binary predictions using τ

$$y_{\text{pred}} \leftarrow (P_{\text{ensemble\_test}} \geq \tau)$$

Output Predictions:

Return y_{pred} for X_{test}

Proposed Solution

Algorithm Ensembling XGB, RF, and ETC for diabetes classifcation

Calculate accuracy, precision, recall, and F1 score for $\hat{y}_{\text{val}}$

Final Predictions on Test Set:

Predict probabilities on X^*_{test} using model_XGB

$$P_{\text{XGB\_test}} \leftarrow \text{model\_XGB.predict\_proba}(X^*_{\text{test}})[:, 1]$$

Predict probabilities on X^*_{test} using model_RF

$$P_{\text{RF\_test}} \leftarrow \text{model\_RF.predict\_proba}(X^*_{\text{test}})[:, 1]$$

Predict probabilities on X^*_{test} using model_ETC

$$P_{\text{ETC\_test}} \leftarrow \text{model\_ETC.predict\_proba}(X^*_{\text{test}})[:, 1]$$

Combine the predictions using the average method

$$P_{\text{ensemble\_test}} \leftarrow \frac{P_{\text{XGB\_test}} + P_{\text{RF\_test}} + P_{\text{ETC\_test}}}{3}$$

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Model Training:

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Model Prediction on Validation Set:

Predict probabilities on X_{val} using model_XGB

$$P_{\text{XGB}} \leftarrow \text{model\_XGB.predict\_proba}(X_{\text{val}})[:, 1]$$

Predict probabilities on X_{val} using model_RF

$$P_{\text{RF}} \leftarrow \text{model\_RF.predict\_proba}(X_{\text{val}})[:, 1]$$

Predict probabilities on X_{val} using model_ETC

$$P_{\text{ETC}} \leftarrow \text{model\_ETC.predict\_proba}(X_{\text{val}})[:, 1]$$

Ensemble Predictions:

Combine the predictions using the average method

$$P_{\text{ensemble}} \leftarrow \frac{P_{\text{XGB}} + P_{\text{RF}} + P_{\text{ETC}}}{3}$$

Determine Optimal Threshold:

Find the optimal threshold τ that maximizes the F1 score on P_{ensemble} and y_{val}

Evaluate Ensemble Model on Validation Set:

Generate binary predictions using τ

$$\hat{y}_{\text{val}} \leftarrow (P_{\text{ensemble}} \geq \tau)$$

Evaluation Parameters

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall(Sensitivity) = \frac{TP}{TP + FN}$$

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

The MCC is calculated using the formula:

$$\mathrm{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

TP : Patient is Diabetic and the prediction also says the same

FP : Patient is Normal But prediction says he/she is Diabetic

TN : Patient is Normal and Prediction also Says the same

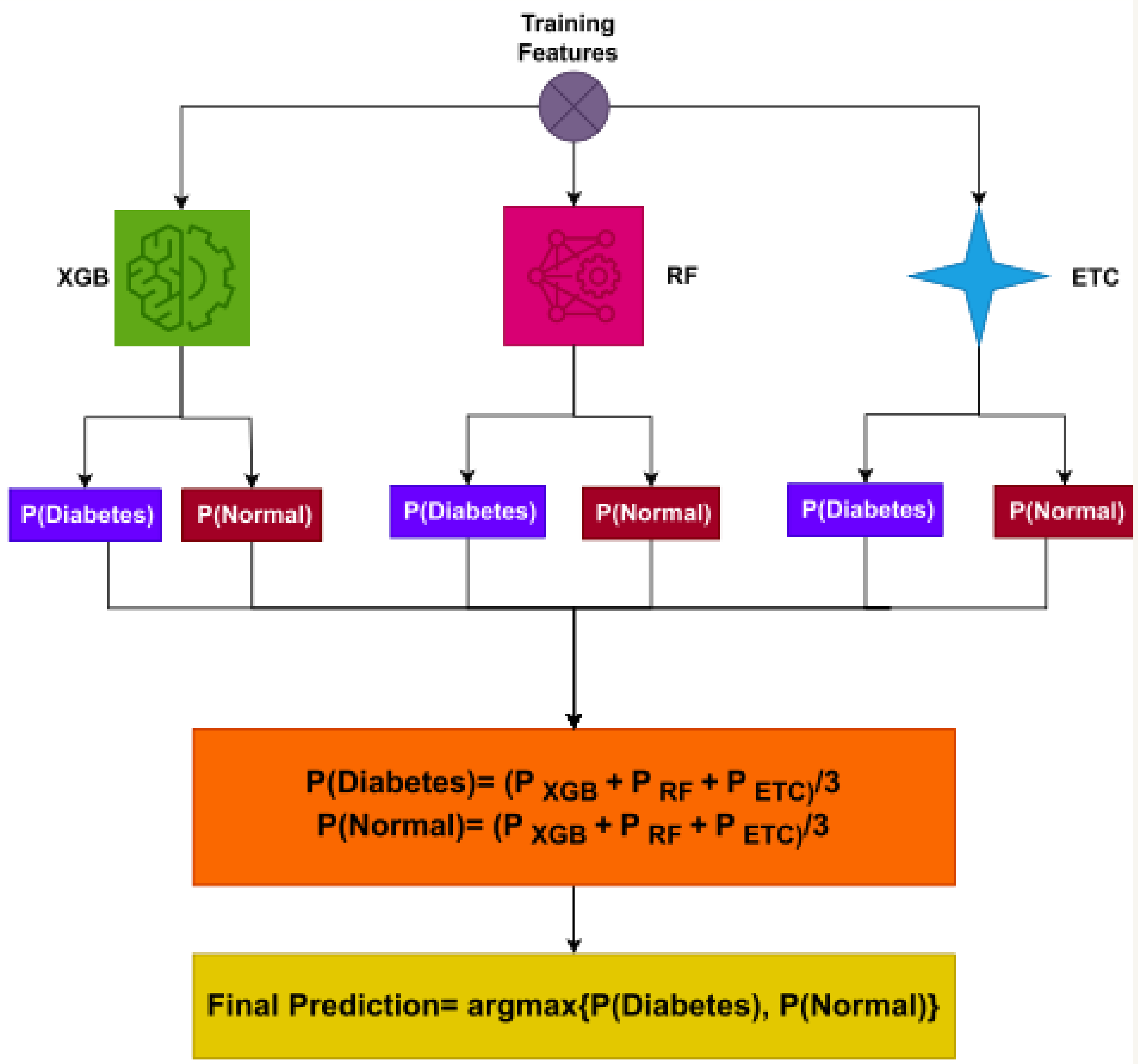
FN : Patient is Diabetic But Prediction Says he/she is Normal

+1 indicates a perfect prediction,

0 indicates no better than random prediction,

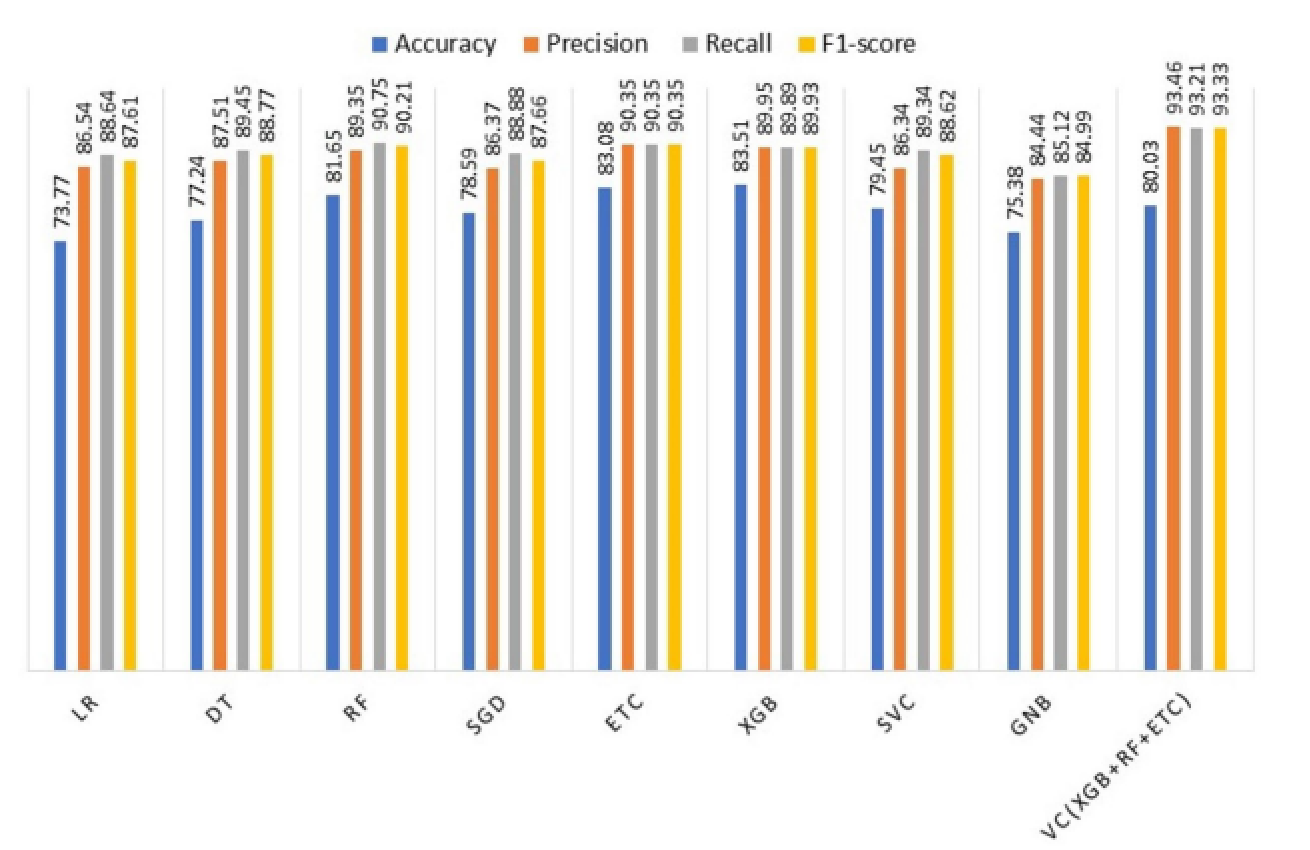
-1 indicates total disagreement between prediction and observation.

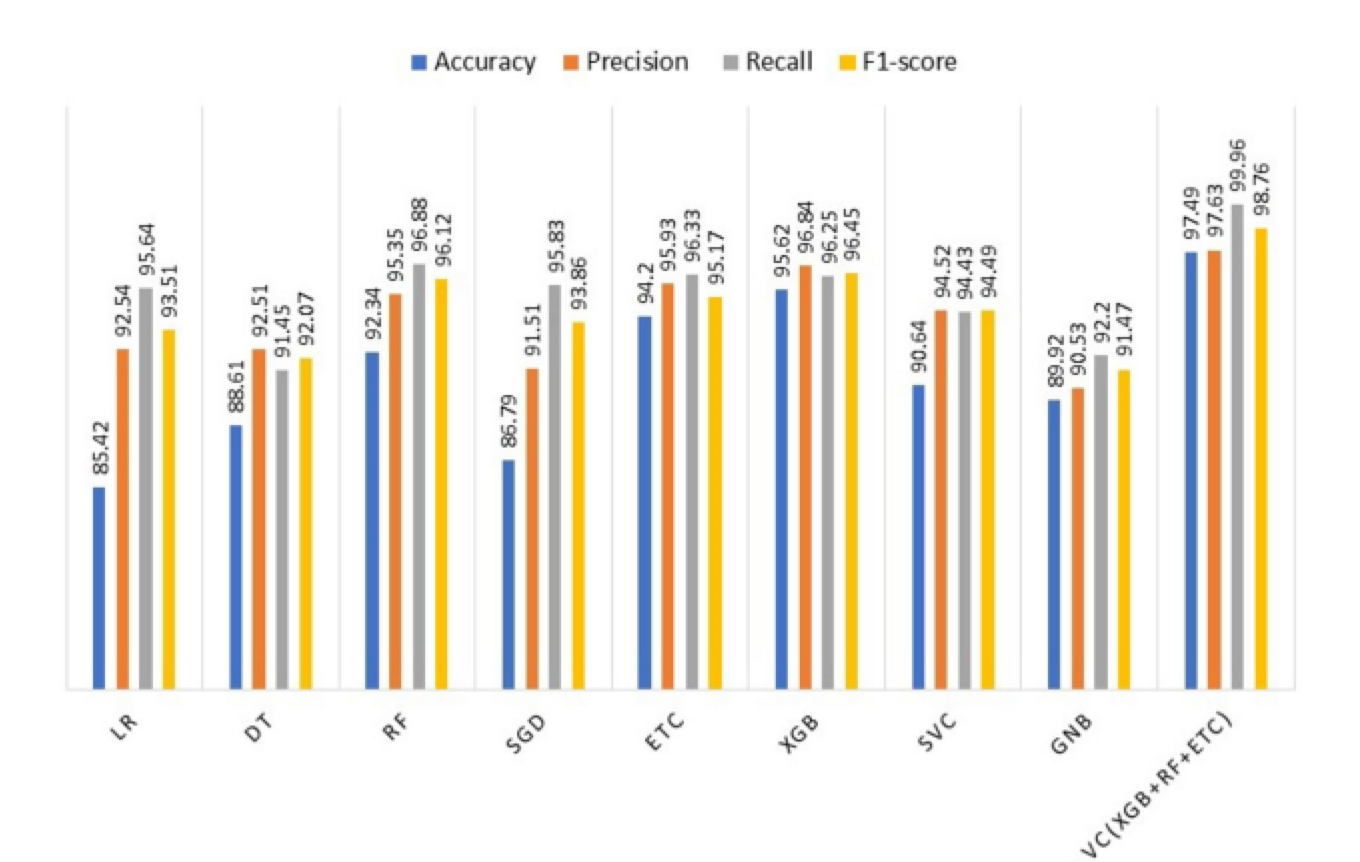
Evaluation Parameters



Results And Discussion

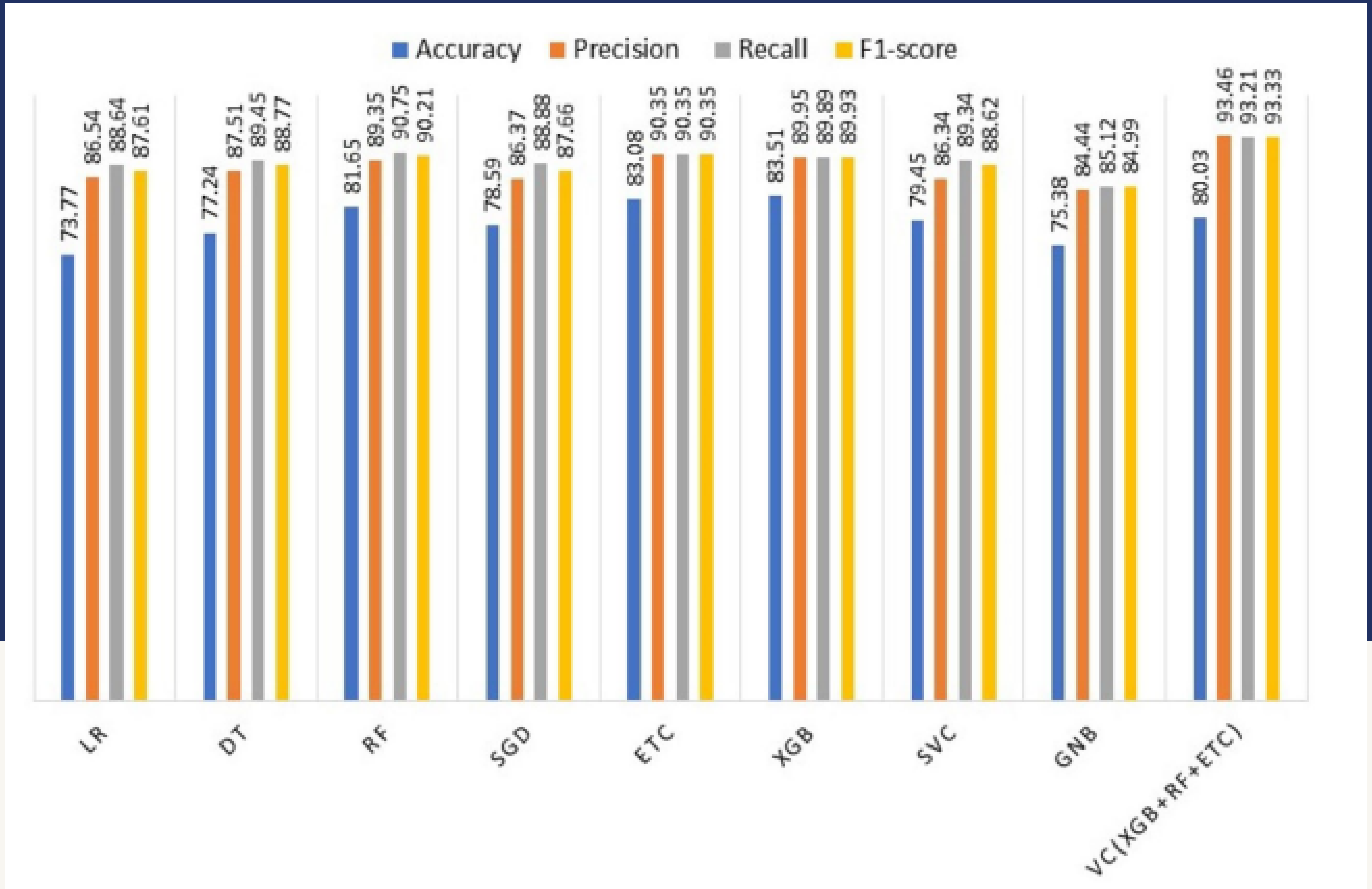
Results of the machine learning models created using variables in the dataset

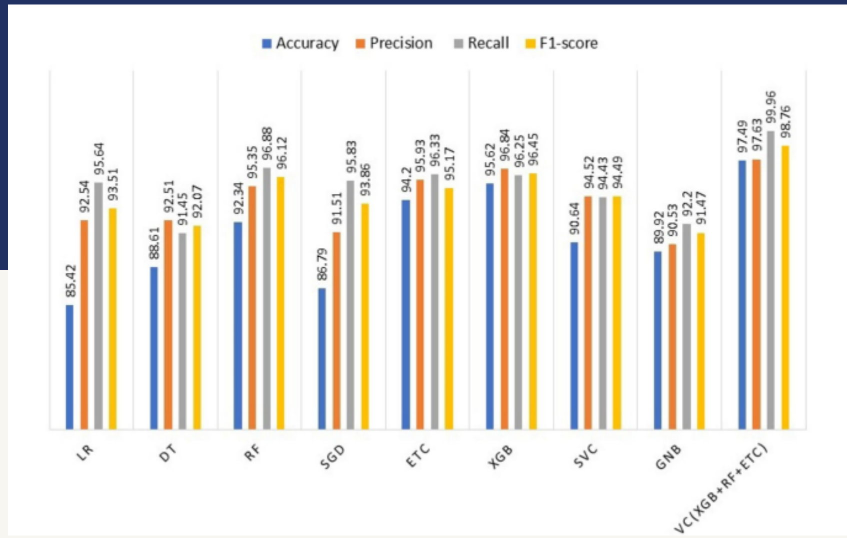




Results And Discussion

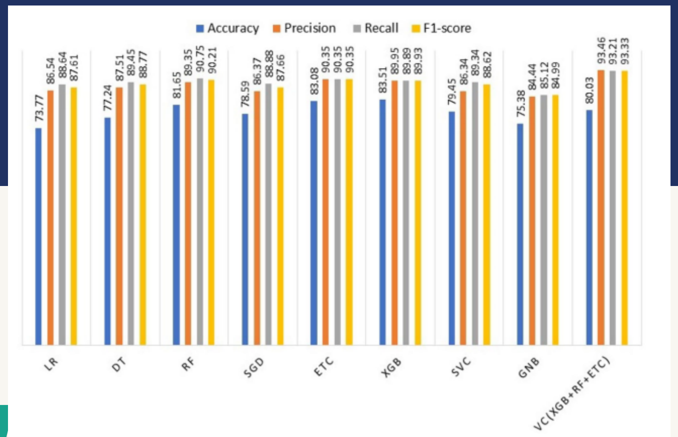
Results of the machine learning models created using the missing variables in the dataset

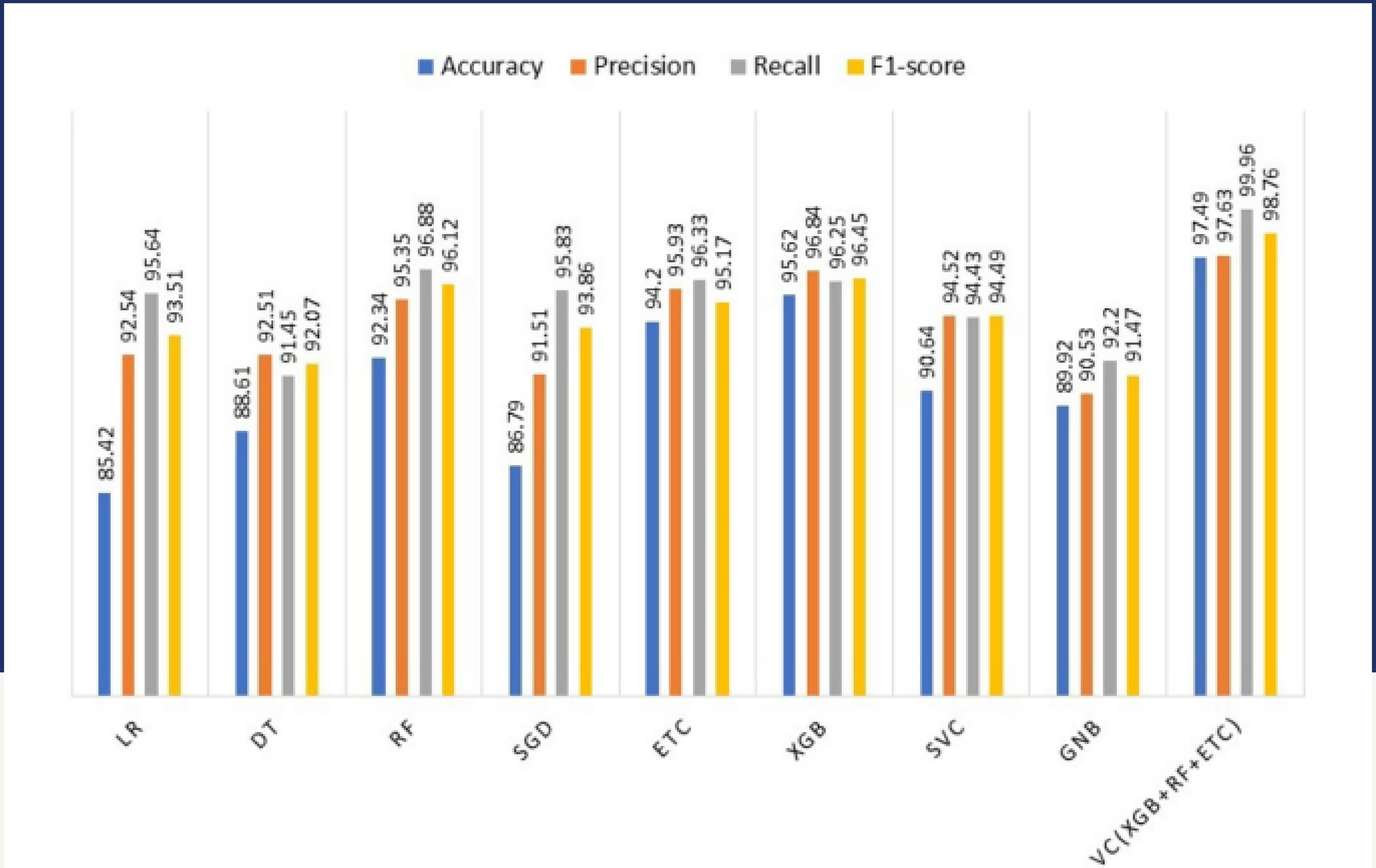




Results And Discussion

Results of the KNN imputer machine learning models





k-fold cross validation

5-fold cross-validation results for the proposed system

Model	Accuracy	Precision	Recall(Sensitivity)	F1-Score
1st fold	97.62	98.23	99.71	99.22
2nd fold	97.35	98.44	99.84	99.33
3rd fold	98.74	99.77	101.08	100.91
4th fold	98.18	99.88	101.09	100.95
5th fold	98.08	98.25	99.96	99.43
Average	**97.99**	**98.91**	**100.33**	**99.97**

Comparison of the Two Experiments

Advantages

Limitations

Conclusion

Comparison of the Two Experiments

Models without KNN imputation exhibited lower accuracy and other performance metrics across the board.

Models with KNN imputation showed marked improvement, with the ensemble model reaching almost perfect accuracy (98.59%) and other metrics.

The KNN imputer helped improve the quality of the data by filling in the missing values, which positively affected the prediction models

Advantages

Improved Accuracy: The ensemble model, after applying KNN imputation, showed significantly higher accuracy and robustness across all metrics.

Effectiveness of KNN Imputation: Handling missing data with KNN imputer significantly boosted performance compared to models without imputation.

Ensemble Learning: Combining models such as XGB, RF, and ETC improved the final prediction accuracy by leveraging the strengths of each individual model

Limitations

Dependence on Data Quality: The proposed model's performance heavily depends on the quality and completeness of input data, as KNN imputation works best with datasets that have minimal and randomly distributed missing values.

KNN Parameter Sensitivity: Performance of the KNN imputer is sensitive to the choice of the number of neighbors (K) and the distance metric used, which requires careful tuning to achieve optimal results

Conclusion

The proposed model combining KNN imputation with ensemble learning (XGB + RF + ETC) demonstrated superior performance over other models and techniques, achieving nearly 99% accuracy. This makes the model highly effective for diabetes prediction tasks where data completeness is a challenge.

Thank You